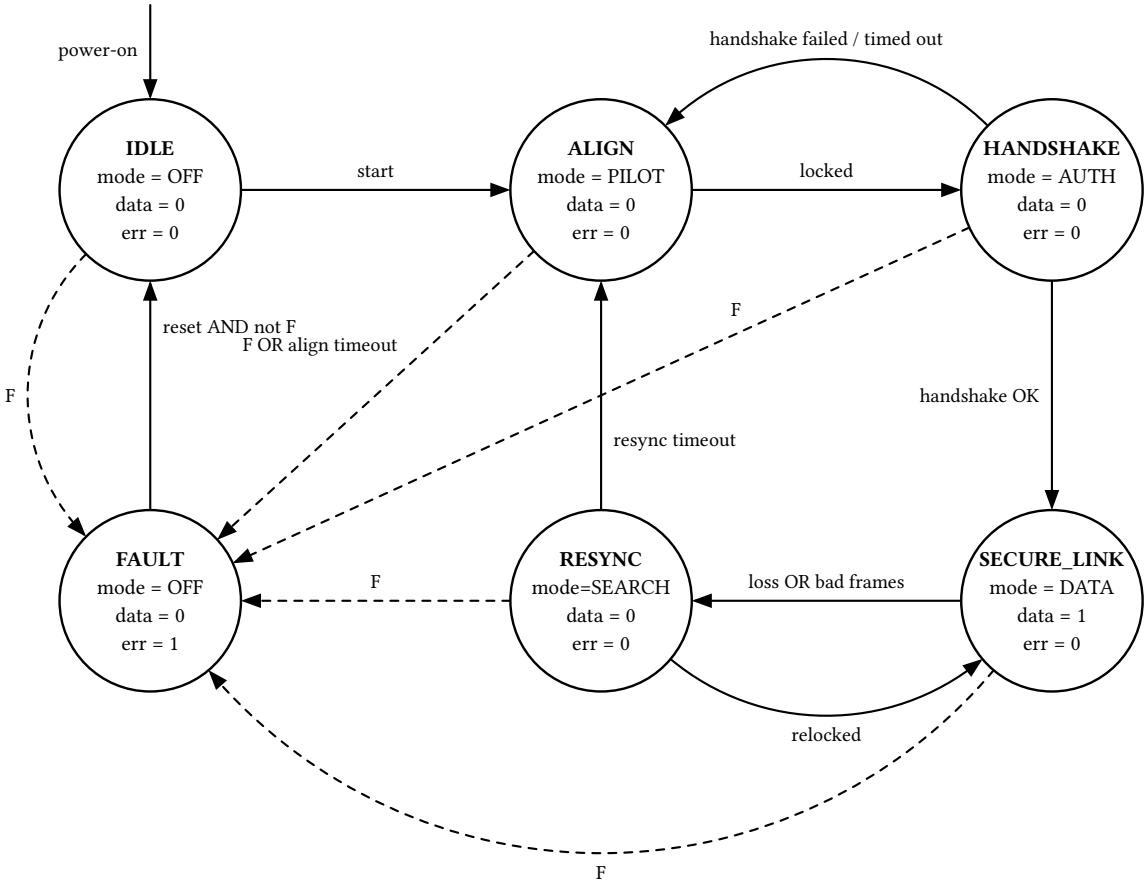

Figure 1: Moore finite state machine for the secure optical link.

Outputs inside each state: `mode` selects the optical datapath operation; `data` enables authenticated payload processing; `err` asserts the fault flag. OFF disables the laser and idles the receiver; PILOT transmits an alignment pattern and searches for the preamble; AUTH enables the handshake; DATA enables secure frames; SEARCH enables preamble reacquisition while retaining the key and counters. These control outputs depend only on the state.

Transition rules: F = hardware error OR authentication-failure limit reached. “loss OR bad frames” means signal lost OR the bad-frame limit reached. Dashed arrows enter FAULT. F has highest priority; all other transitions require not F. Alignment timeout takes priority over locked, handshake failure/timeout over handshake OK, and resync timeout over relocked. If no outgoing condition is true, stay in the current state (implicit self-loop). FAULT exits only on reset with F cleared. Power-on enters IDLE.

Handshake OK means the peer key is confirmed and fresh, non-reused nonce/counter state is established. Individual frame authentication and replay checks are performed by the datapath before plaintext delivery.